D Parity bit generator & checker

Extra bit added to message to make the o/p to be odd or even

① odd parity ② Even parity

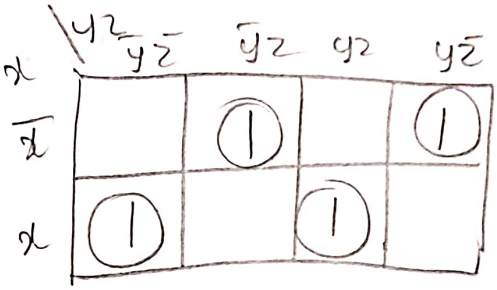
checker
0 → correct
1 → error

- Design and implement even parity generator and checker.

Message

x	y	z	P_{even}
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	0
1	1	1	1

0 → to make odd
1 → " even

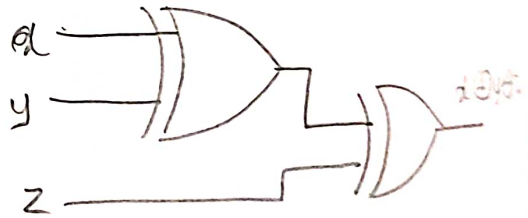


$$\bar{x}\bar{y}z + \bar{x}y\bar{z} + x\bar{y}\bar{z} + xyz$$

$$\bar{x}(\bar{y}z + y\bar{z}) + x(\bar{y}\bar{z} + yz)$$

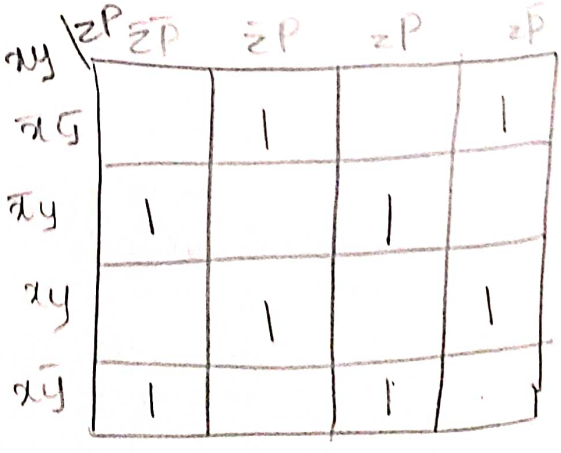
$$\bar{x}(y \oplus z) + x(\overline{y \oplus z})$$

$$x \oplus y \oplus z$$



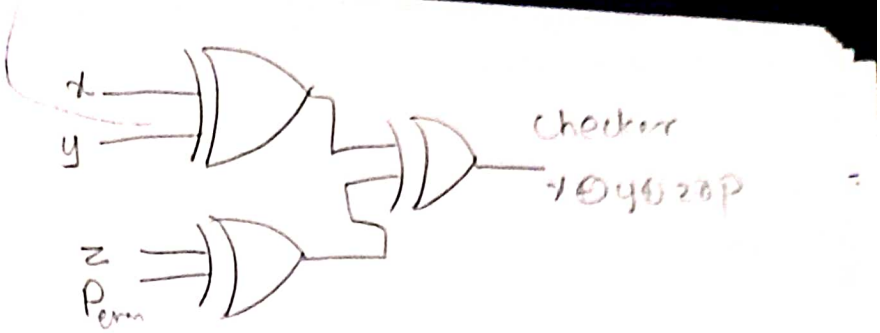
Even parity generator

x	y	z	P_{even}	$checker_{even}$
0	0	0	0	0
0	0	0	1	1
0	0	1	0	1
0	0	1	1	0
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1



checker = $x \oplus y \oplus z \oplus P_{even}$

1	0	0	0
1	0	0	1
1	0	1	0
1	0	1	1
1	1	0	0
1	1	0	1
1	1	1	0
1	1	1	1



X-OR

1. 4-bit adder/subtractor
2. Parity generator & checker

Design and implement odd parity and checker.

x	y	z	Podd
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	0

even

x \ yz	$\bar{y}\bar{z}$	$\bar{y}z$	yz	$y\bar{z}$
$\bar{x}$	1		1	
x		1		1

$\bar{x}\bar{y}\bar{z} + \bar{x}yz + x\bar{y}z + xy\bar{z}$

$\bar{x}(\bar{y}\bar{z} + yz) + x(\bar{y}z + y\bar{z})$

$\bar{x}(\overline{y \oplus z}) + x(y \oplus z)$

$= \overline{x \oplus y \oplus z}$



x	y	z	Podd	$checker_{odd}$
0	0	0	0	1
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	0
0	1	0	1	1
0	1	1	0	1
0	1	1	1	0
1	0	0	0	0
1	0	0	1	1
1	0	1	0	1
1	0	1	1	0
1	1	0	0	1
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

	$z\bar{P}$	$\bar{z}\bar{P}$	$\bar{z}P$	zP	$z\bar{P}$
xy	(1)		(1)		
$x\bar{y}$		(1)		(1)	
$\bar{x}y$	(1)		(1)		
$\bar{x}\bar{y}$		(1)		(1)	

$$\bar{x}\bar{y}\bar{z}\bar{P} + \bar{x}\bar{y}zP + \bar{x}y\bar{z}P + \bar{x}yz\bar{P} + xy\bar{z}\bar{P} + xyzP + x\bar{y}\bar{z}P + x\bar{y}z\bar{P}$$

$$= \bar{x}\bar{y}(\bar{z}\bar{P} + zP) + \bar{x}y(\bar{z}P + z\bar{P}) + xy(\bar{z}\bar{P} + zP) + x\bar{y}(\bar{z}P + z\bar{P})$$

$$= \bar{x}\bar{y}(\overline{z \oplus P}) + \bar{x}y(z \oplus P) + xy(\overline{z \oplus P}) + x\bar{y}(z \oplus P)$$

let $u = z \oplus P$

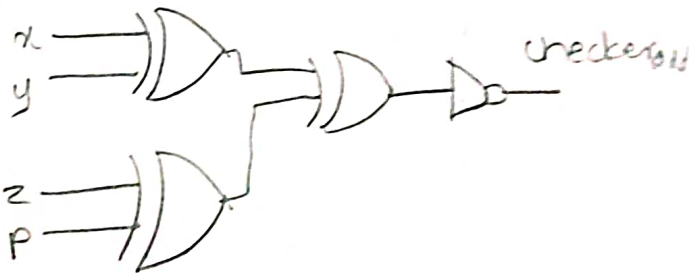
$$= \bar{x}\bar{y}\bar{u} + \bar{x}yu + xy\bar{u} + x\bar{y}u$$

$$= \bar{x}(\bar{y}\bar{u} + yu) + x(y\bar{u} + \bar{y}u)$$

$$= \bar{x}(\overline{y \oplus u}) + x(y \oplus u)$$

$$= \overline{\bar{x} \oplus y \oplus u}$$

$$= x \oplus y \oplus z \oplus P$$



Multiplexer -> read

Implement $u(a,b,c,d) = ad + b\bar{c} + bd$ using 16:1, 8:1 and 4:1 MUX.

$$u(a,b,c,d) = ad + b\bar{c} + bd$$

$$= a(b+\bar{b})(c+\bar{c})d + (a+\bar{a})b\bar{c}(d+\bar{d}) + (a+\bar{a})b(c+\bar{c})d$$

$$= (ab+a\bar{b})(cd+\bar{c}d) + (ab\bar{c} + \bar{a}b\bar{c})(d+\bar{d}) + (ab+\bar{a}b)(cd+\bar{c}$$

$$= abcd + ab\bar{c}d + a\bar{b}cd + a\bar{b}\bar{c}d + ab\bar{c}d + ab\bar{c}\bar{d} + \bar{a}b\bar{c}d + \bar{a}b\bar{c}\bar{d}$$

(15) (13) (11) (9) (13) (12) (5) (4)

$abcd + ab\bar{c}d + \bar{a}bcd + \bar{a}b\bar{c}d$

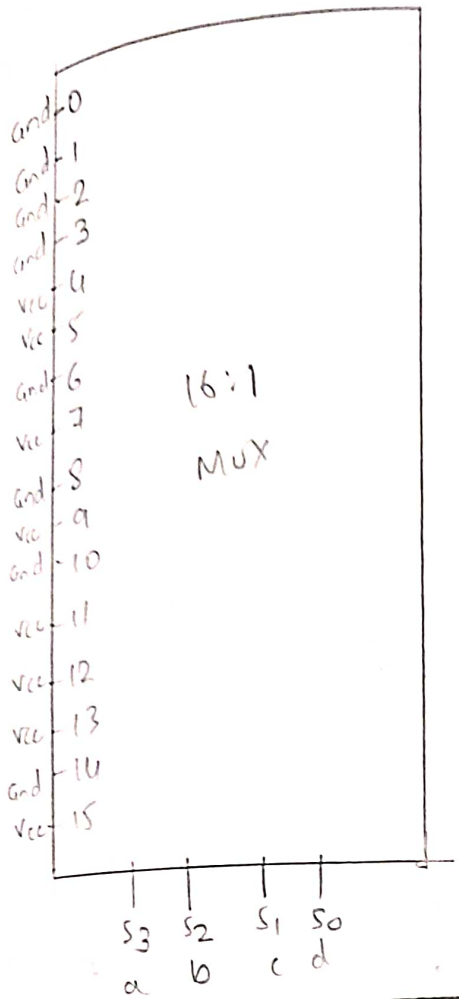
(15) (13) (7) (5)

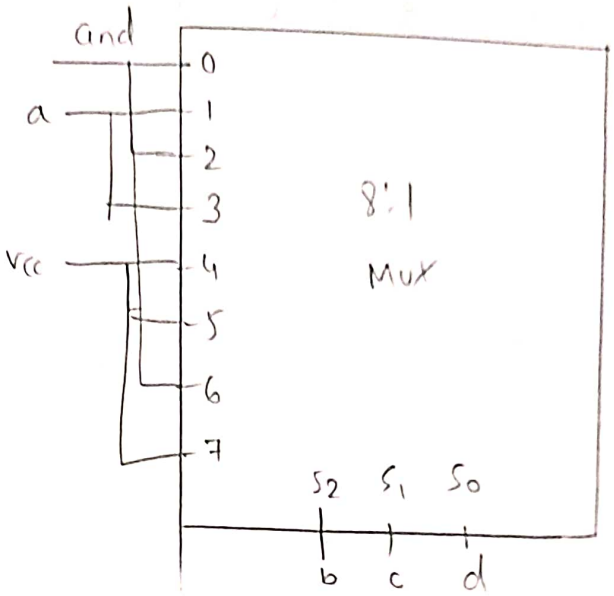
$$= \Sigma m(4, 5, 7, 9, 11, 12, 13, 15)$$

16:1 $16 = 2^{(4)}$ → selection lines

$\bar{a}$	0	1	2	3	(4)	(5)	6	(7)
a	8	(9)	(10)	(11)	(12)	(13)	14	(15)

Gnd d Gnd d V_{cc} V_{cc} Gnd V_{cc} (3)

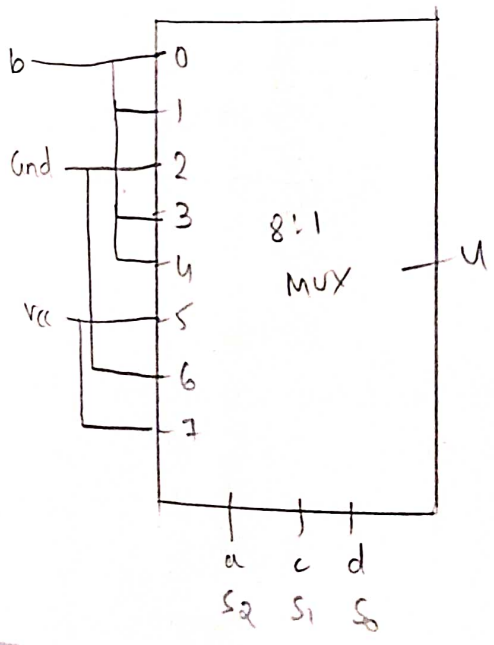




8:1 MUX → b to be i/p a c d selection line

$\bar{b}$	0	1	2	3	8	(9)	10	(11)
b	(6)	(5)	6	(7)	(2)	(3)	14	(15)

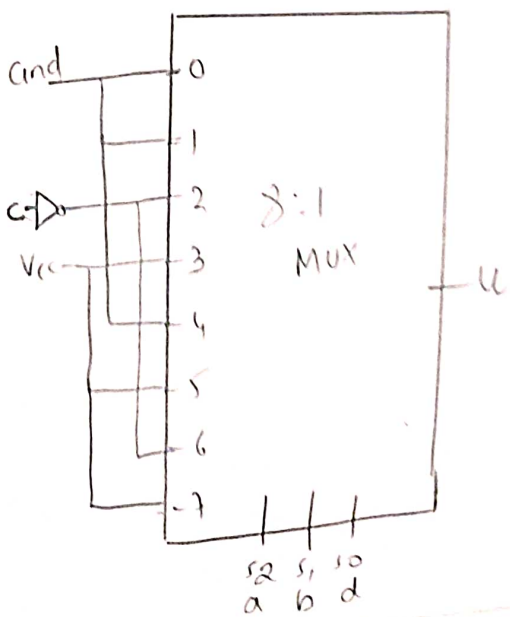
b b Gnd b b Vcc Gnd Vcc



c as i/p a b d selection line

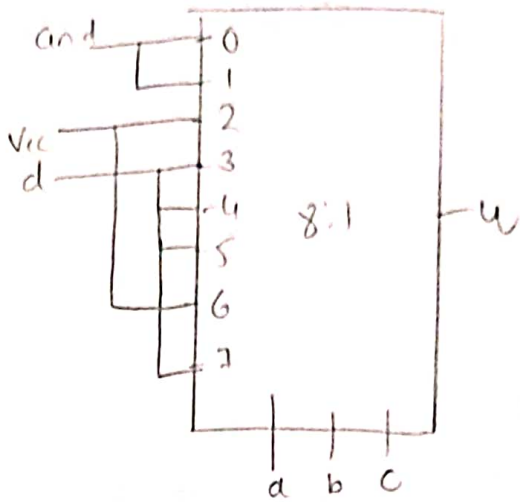
$\bar{c}$	0	1	(4)	(5)	8	(9)	(12)	(13)
c	2	3	6	(7)	10	(11)	14	(15)

Gnd Gnd $\bar{c}$ Vcc Gnd Vcc $\bar{c}$ Vcc



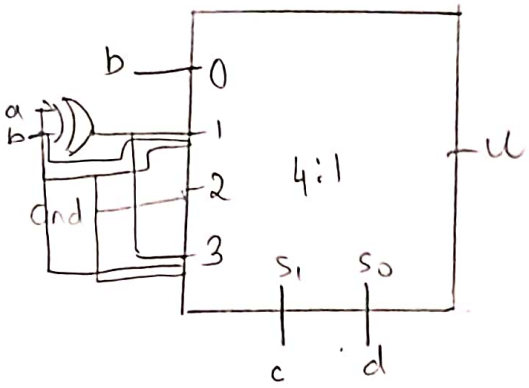
d as i/p and a, b, c as selection lines

$\bar{d}$	0	2	(4)	6	8	10	(12)	14
d	1	3	(5)	(7)	(9)	(11)	(13)	(15)
	0	0	V_{cc} d	d	d	V_{cc}	d	



4:1

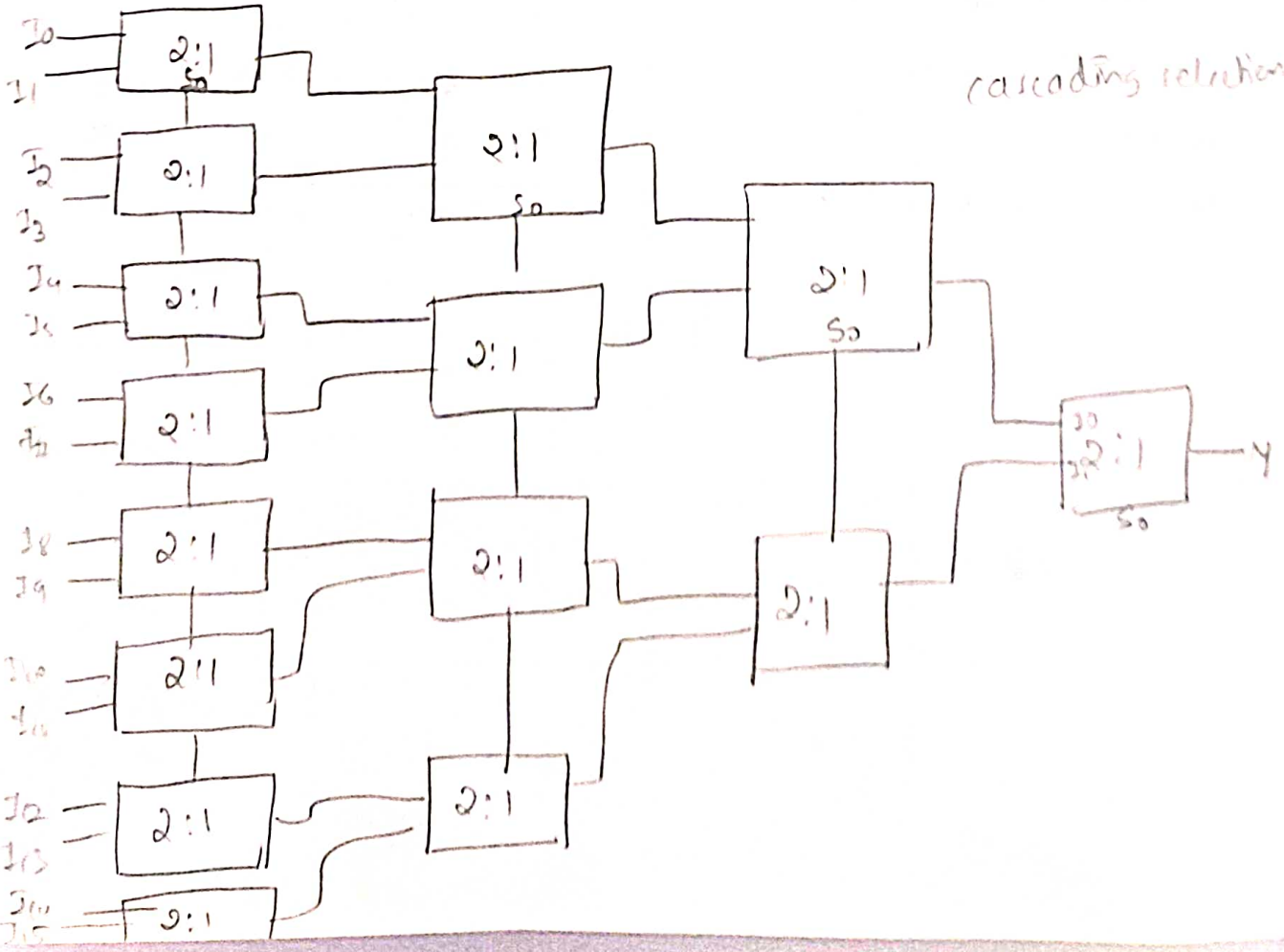
$\bar{a}\bar{b}$	0	1	2	3
$\bar{a}b$	(4)	(5)	6	(7)
$a\bar{b}$	8	(9)	10	(11)
ab	(12)	(13)	14	(15)
	b	$\bar{a}b$ + $a\bar{b}$ + a+b	and	$\bar{a}b$+$a\bar{b}$ + a+b
	I_0	I_1	I_2	I_3



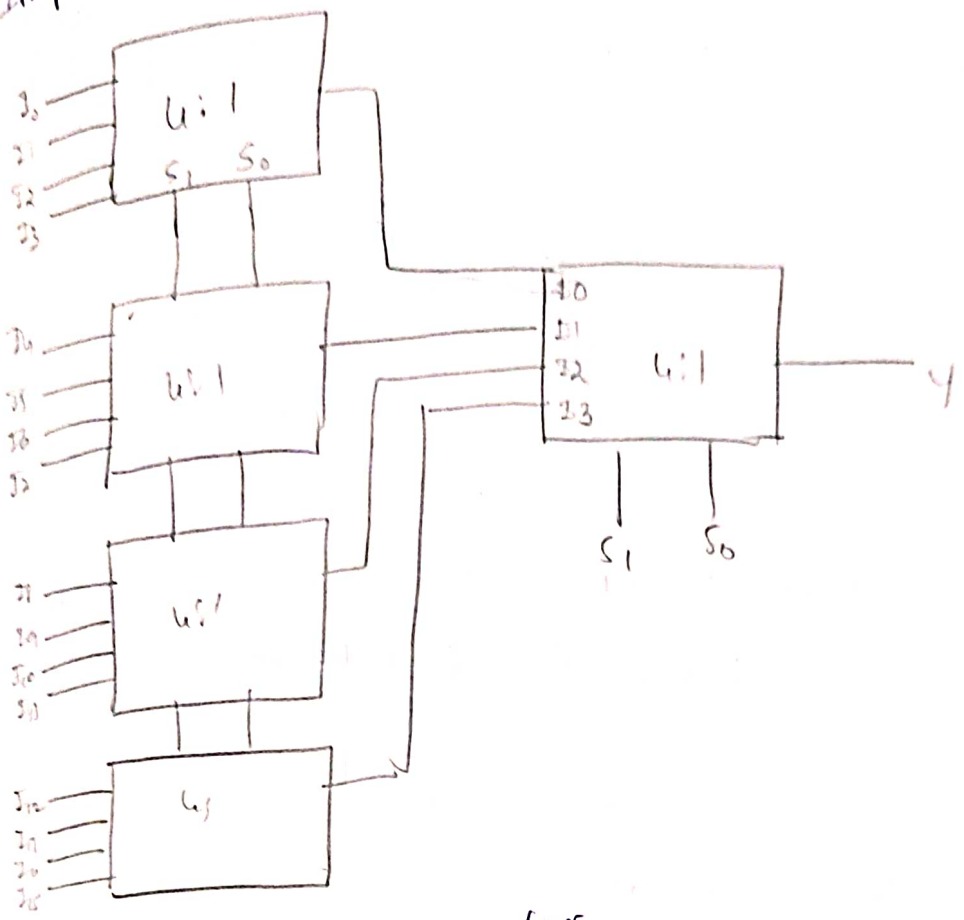
Implement 16:1 MUX using 2:1 MUX.

Enable always to gnd

cascading selection lines

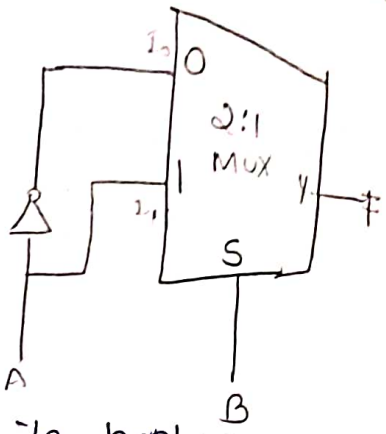


Implement 16:1 MUX using 4:1 MUX



Configure 13:1 MUX using four 8:1 MUX.

1) Consider the following circuit which uses a 2:1 MUX as shown in the figure. Find the boolean expression for o/p y in terms of a & b.

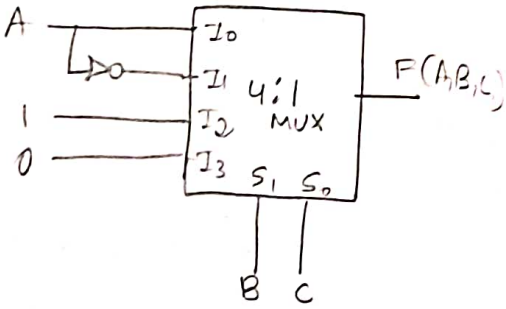


$F = \bar{S} I_0 + S I_1$

$F = \bar{B}\bar{A} + BA = \overline{A\bar{B}} + AB$

$y = \overline{A \oplus B} = A \odot B$

2) A 4:1 MUX is used to implement a 3 i/p boolean function as shown in fig. Find the boolean fn $F(A, B, C)$



$Y = \overset{0\,0}{\bar{S}_1\bar{S}_0} I_0 + \overset{0\,1}{\bar{S}_1 S_0} I_1 + \overset{1\,0}{S_1\bar{S}_0} I_2 + \overset{1\,1}{S_1 S_0} I_3$

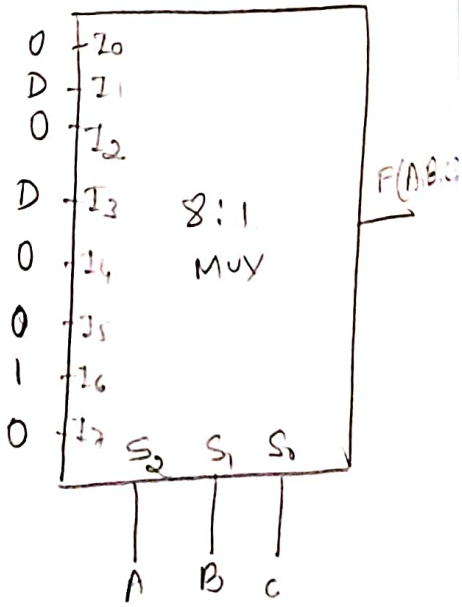
$= \bar{B}\bar{C}A + \bar{B}C\bar{A} + B\bar{C}.1 + BC.(0)$

$= \bar{B}\bar{C}A + A\bar{B}\bar{C} + \bar{A}\bar{B}C + B\bar{C}(A+\bar{A})$

$= \underset{4}{A\bar{B}\bar{C}} + \underset{1}{\bar{A}\bar{B}C} + \underset{6}{AB\bar{C}} + \underset{2}{\bar{A}B\bar{C}}$

$= \Sigma m(1,2,4,6)$

3) An 8:1 MUX is used to implement a logical fn y as shown fig. Find the o/p y in boolean expression.



$F = \bar{S}_2\bar{S}_1\bar{S}_0 I_0 + \bar{S}_2\bar{S}_1 S_0 I_1 + \bar{S}_2 S_1\bar{S}_0 I_2 +$

$\bar{S}_2 S_1 S_0 I_3 + S_2\bar{S}_1\bar{S}_0 I_4 + S_2\bar{S}_1 S_0 I_5 + S_2 S_1\bar{S}_0 I_6$

$+ S_2 S_1 S_0 I_7$

$= \bar{A}\bar{B}\bar{C}(0) + \bar{A}\bar{B}C(D) + \bar{A}B\bar{C}(0) + \bar{A}BCD +$

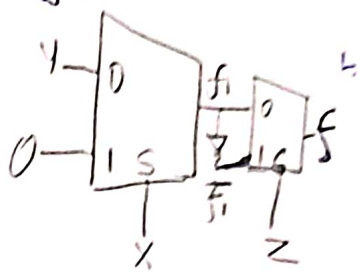
$A\bar{B}\bar{C}(0) + A\bar{B}C(0) + AB\bar{C}(1) + ABC(0)$

$= \bar{A}BCD + \bar{A}BCD + AB\bar{C}$

$= \bar{A}CD(\bar{B}+B) + AB\bar{C}$

$= \bar{A}CD + AB\bar{C}$

4) Consider the circuit shown below. Find boolean exp^n 'f'.



$f_1 = \bar{S} I_0 + S I_1$

$= \bar{X} Y + X(0)$

$= \underline{\bar{X} Y}$

$\overline{\bar{X} Y} = \bar{\bar{X}} + \bar{Y}$

$= X + \bar{Y}$

$f = \bar{S} f_1 + S \bar{F_1}$

$= \bar{Z} \bar{X} Y + Z(\overline{\bar{X} Y})$

$= \bar{X} Y \bar{Z} + Z(X + \bar{Y}) = \bar{X} Y \bar{Z} + XZ + \bar{Y} Z$

5) Implement the fn $f = A \odot B \oplus C$ using 8:1 MUX and 4:1 MUX.

$(A \odot B) \oplus C$

$[\bar{A}\bar{B} + AB] \oplus C$

$(\overline{\bar{A}\bar{B} + AB} C + (\bar{A}B + A\bar{B})\bar{C}$

$(\bar{A}\bar{B} \cdot \overline{AB}) C + \bar{A}\bar{B}\bar{C} + AB\bar{C}$

$(A+B) \cdot (\bar{A} + \bar{B}) C + \bar{A}\bar{B}\bar{C} + AB\bar{C}$

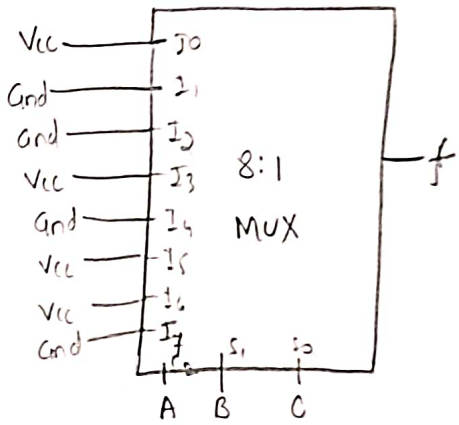
$(A+B)(\bar{A}C + \bar{B}C) + \;''$

$A\bar{B}C + A\bar{B}C + \bar{A}BC + B\bar{B}C + \bar{A}\bar{B}\bar{C} + AB\bar{C}$

$A\bar{B}C + \bar{A}BC + \bar{A}\bar{B}\bar{C} + AB\bar{C}$

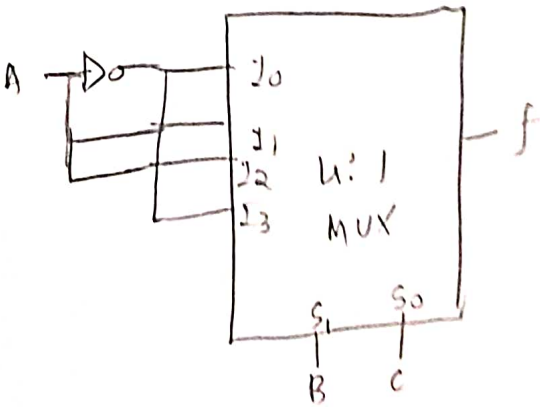
5 3 0 6

$\sum_m (0, 3, 5, 6)$



A	B	C	f
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	0
1	1	0	1
1	1	1	0

$\bar{A}$	0	1	2	3
A	4	5	6	7
	$\bar{A}$	A	A	$\bar{A}$
	I_0	I_1	I_2	I_3



Imp

Decoder:- combinational circuit that takes n i/p and gives 2^n o/p.

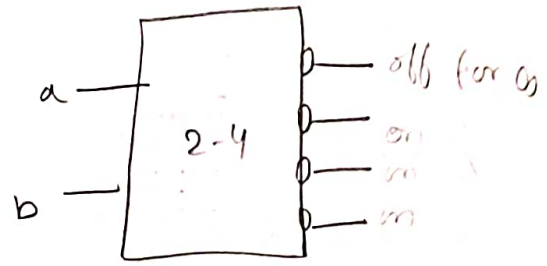
2-4 ✓ 74139

3-8 ✓ 74138

Encoder → $2^n - n$

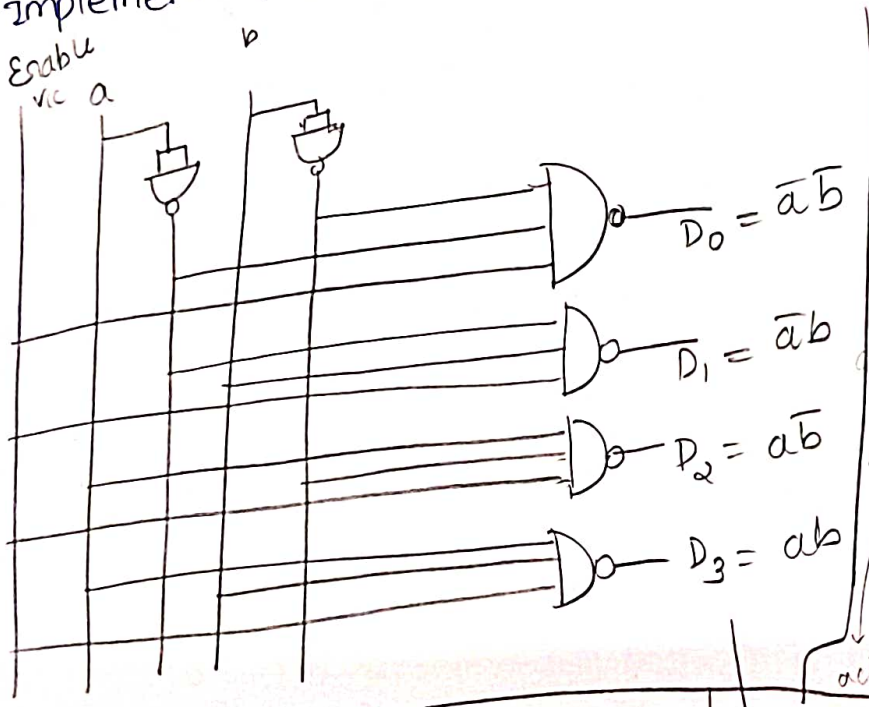
Active low o/p → AND gate

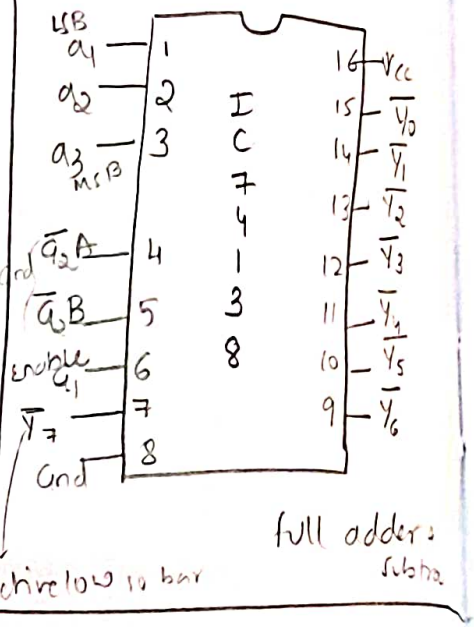
• high " → NAND "



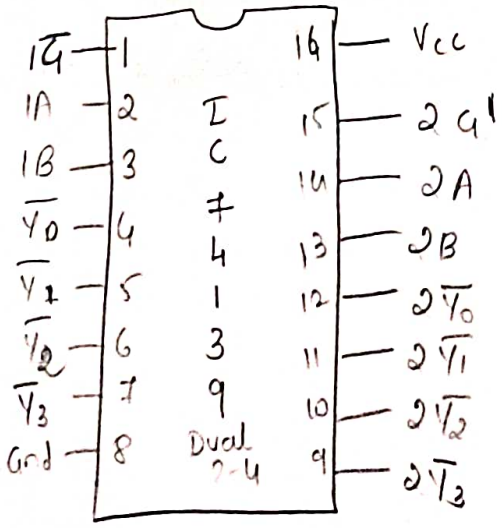
Enable → high (1) → Vcc

Implement 2-4 decoder using only NAND gates:





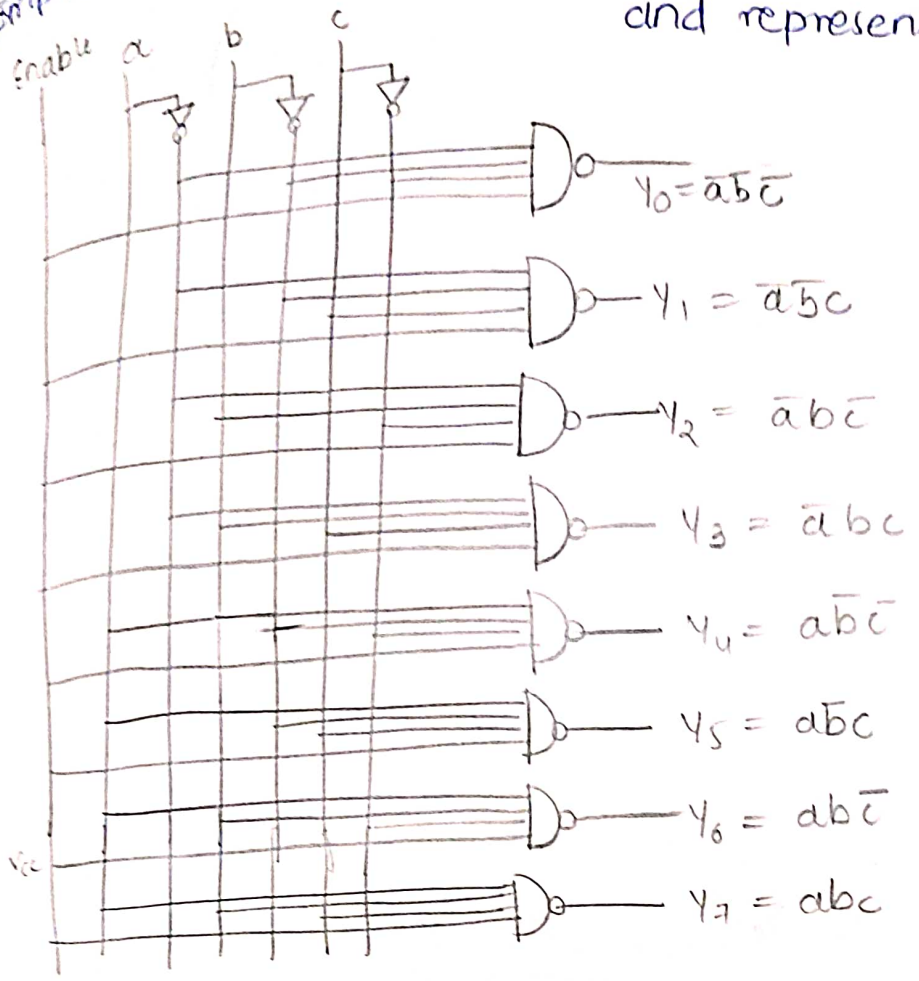
→ half adder & subb



Truth table

I/p			O/p			
E	A	B	$\bar{Y}_0$	$\bar{Y}_1$	$\bar{Y}_2$	$\bar{Y}_3$
1	0	0	0	1	1	1
1	0	1	1	0	1	1
1	1	0	1	1	0	1
1	1	1	1	1	1	0

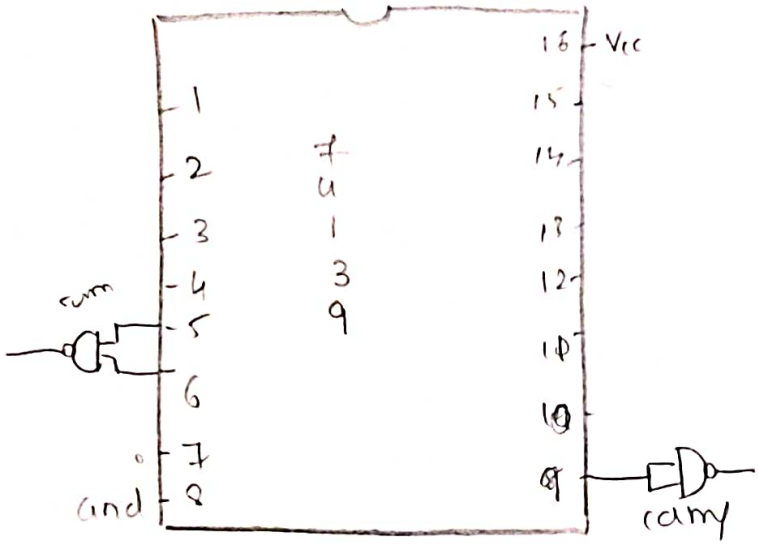
2. Implement 3-8 decoder using ~~only~~ NOT and NAND gates. and represent truth table of the circuit.



E	a	b	c	Y_0	Y_1	Y_2	Y_3	Y_4	Y_5	Y_6	Y_7
1	0	0	0	0	1	1	1	1	1	1	1
1	0	0	1	1	0	1	1	1	1	1	1
1	0	1	0	1	1	0	1	1	1	1	1
1	0	1	1	1	1	1	0	1	1	1	1
1	1	0	0	1	1	1	1	0	1	1	1
1	1	0	1	1	1	1	1	1	0	1	1
1	1	1	0	1	1	1	1	1	1	0	1
1	1	1	1	1	1	1	1	1	1	1	0

3) Implement half adder and half subtractor using dual 2-4 decoder (IC-74139).

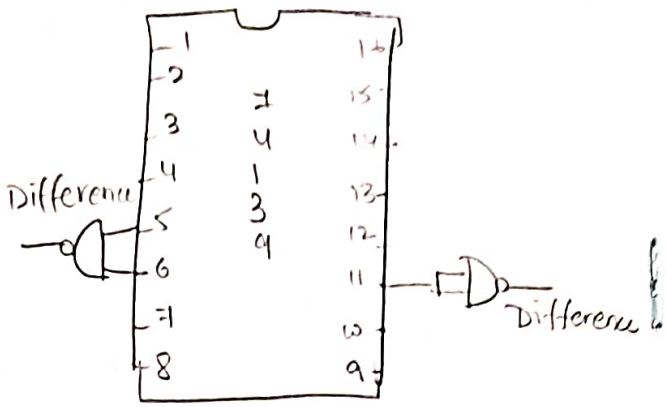
Half adder



A	B	S	C
0	0	0	0
0	1	1	0
1	0	1	0
1	1	0	1

Half subtractor

A	B	D	
0	0	0	
0	1	1	
1	0	0	
1	1	0	

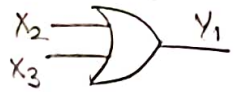


Encoders $2^n \rightarrow n$

4:2 encoder

X_0	X_1	X_2	X_3	Y_1 (MSB)	Y_0 (LSB)
1	0	0	0	0	0
0	1	0	0	0	1
0	0	1	0	1	0
0	0	0	1	1	1

$Y_1 = X_2 + X_3$



$Y_0 = X_1 + X_3$



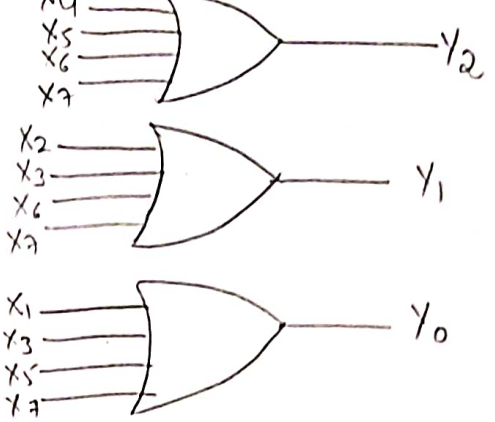
- Implement 8:3 encoder by representing truth table, boulean expression and logic diagram.

X_0	X_1	X_2	X_3	X_4	X_5	X_6	X_7	Y_2	Y_1	Y_0
1	0	0	0	0	0	0	0	0	0	0
0	1	0	0	0	0	0	0	0	0	1
0	0	1	0	0	0	0	0	0	1	0
0	0	0	1	0	0	0	0	0	1	1
0	0	0	0	1	0	0	0	1	0	0
0	0	0	0	0	1	0	0	1	0	1
0	0	0	0	0	0	1	0	1	1	0
0	0	0	0	0	0	0	1	1	1	1

$Y_2 = X_4 + X_5 + X_6 + X_7$

$Y_1 = X_2 + X_3 + X_6 + X_7$

$Y_0 = X_1 + X_3 + X_5 + X_7$



* Priority encoder

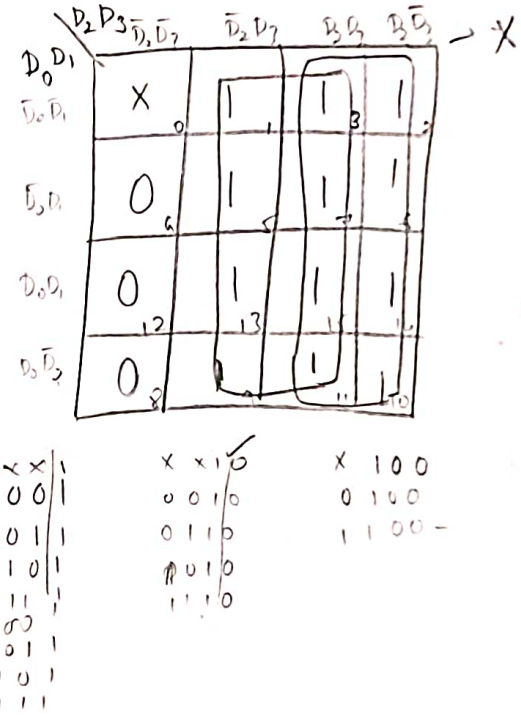
Design a priority encoder for the truth table shown below. Assume that D_3 has the highest priority and D_0 has the lowest priority.

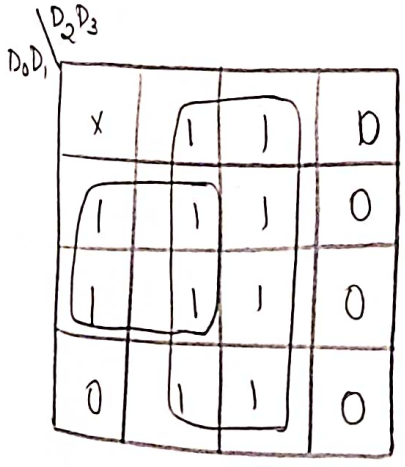
I/P				O/P		
			highest priority ↓			validity bit ↓
D_0	D_1	D_2	D_3	X	Y	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1

lowest priority

0 1 0 0

$X = D_3 + D_2$





x 1 0 0 → 1

0 1 0 0

1 1 0 0

x x 1 0 → 0

0 0 1 0

0 1 1 0

1 0 1 0

1 1 1 0

x x x 1

0 0 0 1

0 0 1 1

0 1 0 1

0 1 1 1

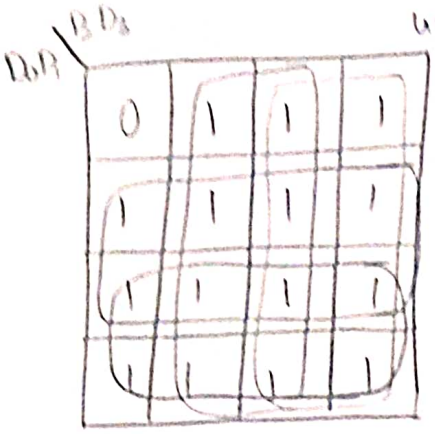
1 0 0 1

1 0 1 1

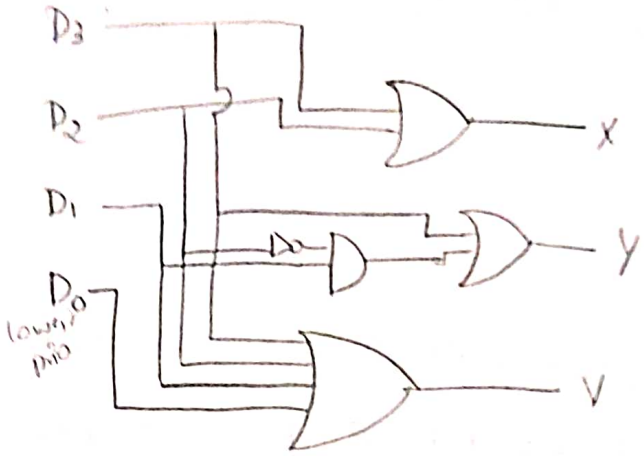
1 1 0 1

1 1 1 1

$Y = D_3 + D_1\overline{D_2}$

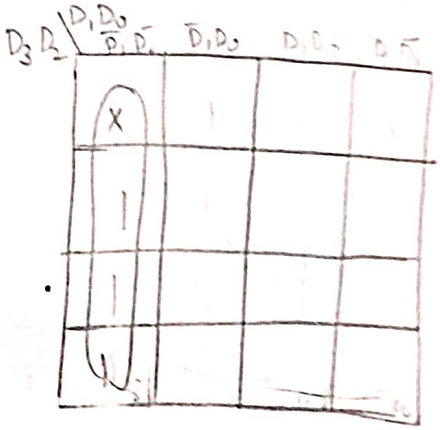


$V = D_0 + D_1 + D_2 + D_3$

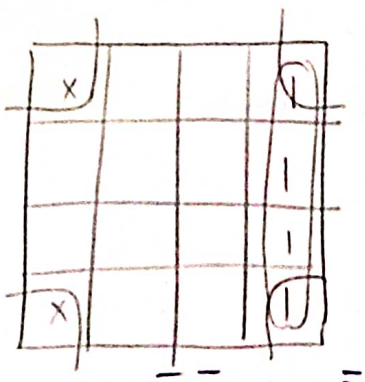


Design a 4-i/p priority encoder with i/p and o/p described in the table below. Note that D_0 has highest priority and D_3 has the lowest priority.

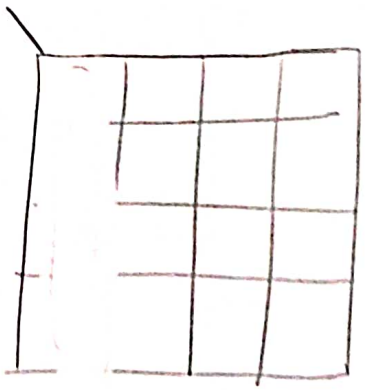
D_3	D_2	D_1	D_0	x	y	V
0	0	0	0	x	x	0
1	0	0	0	0	0	1
x	1	0	0	0	1	1
x	x	1	0	1	0	1
x	x	x	1	1	1	1



$X = \overline{D_1}\,\overline{D_0}$



$Y = \overline{D_2}\,\overline{D_0} + D_1\overline{D_0}$



$V = D_0 + D_1 + D_2 + D_3$